Joining Processes>>

Welding > Introduction>

- **Welding is a process** of joining two pieces of metal by the application of heat with or without the application of pressure and filler material.
- **Welding produces** a permanent joint.
- **Welding is used** in joining pressure vessels, tanks, bridges, railways, machine frames & brackets, building the body of automobiles, aircrafts & ships etc.

Welding >> Classification>

Welding process may be broadly classified as

1. Plastic welding or Pressure welding
2. Fusion welding or Non Pressure welding

1. Plastic welding or Pressure welding

- **In this type of welding,** the metal pieces to be joined are heated to a plastic state and then joined together by the application of pressure without the addition of filler material.
- Ex: Forge welding, Resistance welding.

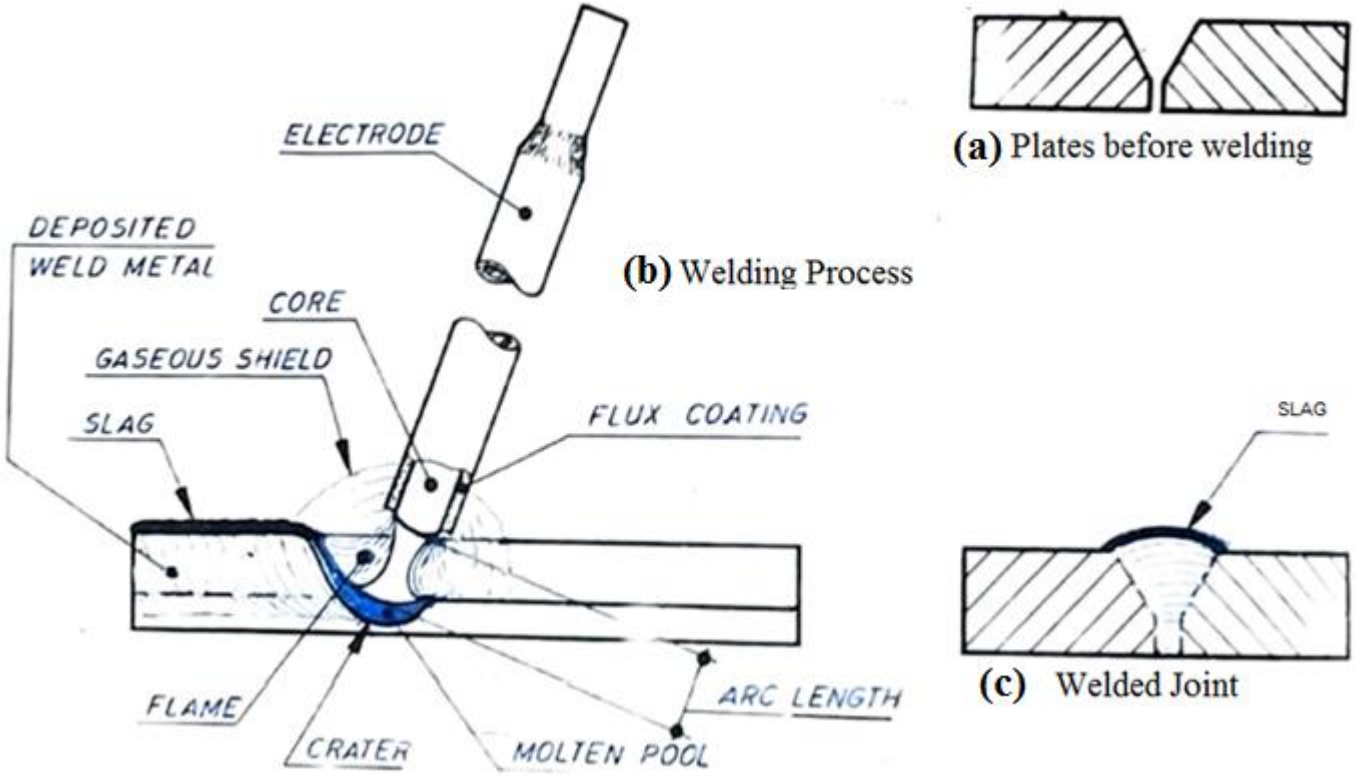
2. Fusion welding or Non Pressure welding

- **In this type of welding,** the metal pieces to be joined are heated to molten state and allowed to solidify without the application of pressure. A filler material is used during the welding process.
- Ex: Arc welding, Gas welding.

Joining Processes>>

Arc Welding Process>

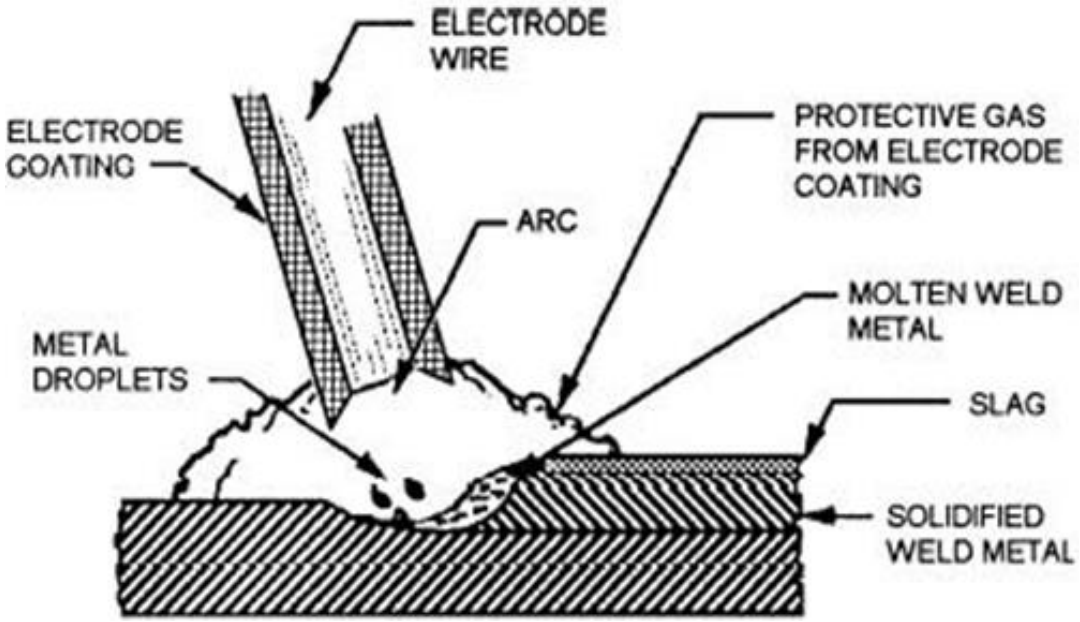
- The principle of arc welding is that, when two conductors of an electric circuit are touched together momentarily and then instantaneously separated slightly, an electric arc is formed.
- A high heat density is produced throughout the length of the arc at a temperature of 5000 to 6000 Centigrade.
- The parts to be welded are wired as one pole of the circuit, and the electrode held by the operator forms the other pole.
- When the arc is produced, the intense heat quickly melts the work piece metal which is directly under the arc, forming a small molten metal pool.
- At the same time the tip of the electrode at the arc also melts, and this molten metal of the electrode is carried over by the arc to the molten metal pool of the workpiece.
- The molten metal in the pool is agitated by the action of the arc, thoroughly mixing the base and the filler metal. A solid joint will be formed when the molten metal cools and solidifies.
- The flux coating over the electrode produces an inert gaseous shield surrounding the arc and protects the molten metal from oxidizing by coming in contact with the atmosphere.

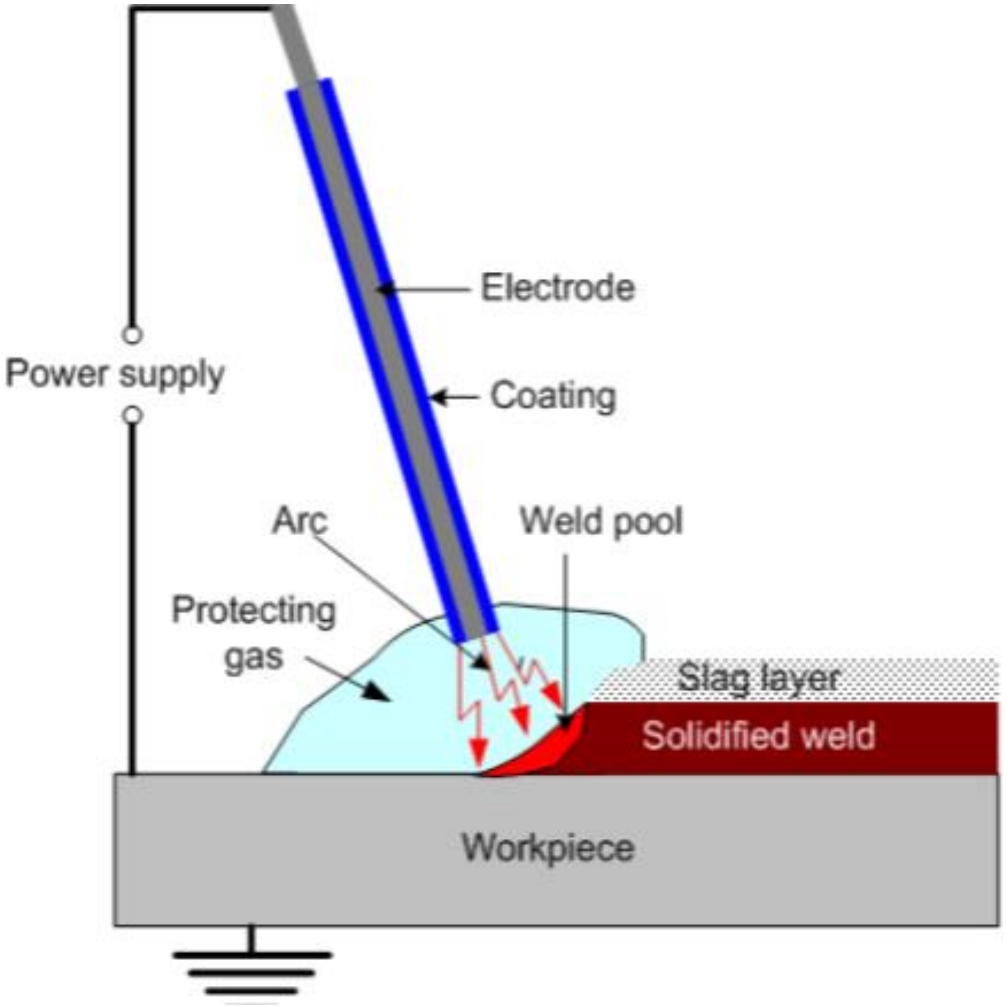


Shielded Metal Arc Welding (SMAW)

Joining Processes>>

Arc Welding Process>





Schematics of Shielded Metal Arc Welding (SMAW) Process and weld Zone

Joining Processes>>

Arc Welding Machine>>

- **The function of a welding machine** is to generate a low voltage (10 to 50 V) and high current (50 A to 300 A).
- **The current may be alternating current or direct current** and the polarity of the electrode may be positive or negative, depending on the type of electrode and the metals to be welded.
- **In AC arc welding**, a step down transformer is used to step down the voltage from 220/440 V to 80 to 100 V & a current of 100 to 400 A.
- **In DC welding,** the work piece is connected to the +ve pole of a D.C generator and the electrode to –ve pole. It is called 'straight polarity' and is used when high heat is required.
- **When less heat is required**, the polarity is reversed.
- **Because of the option of reversing the polarity,** D.C welding may be used to weld many metals which require more heat to melt.
- **In AC welding,** the polarity changes in every cycle.
- **Also the current & the voltage** acquire a value of zero twice in every cycle, and hence higher voltage is required to maintain the arc.

Arc Welding Machine

Joining Processes>>

Arc Welding Electrodes>>

- **The electrode** used in arc welding is of consumable type & it is coated with flux.
- **The flux coating** is usually made of chalk (lime), Ferro manganese, cellulose, Starch, Kaolin (China clay), iron powder, etc.
- **The flux forms** a slag after welding which can be removed by chipping hammer & a wire brush.
- **The purpose of coating** the electrodes are;
  - Protection of molten metal from oxidation
  - To prevent rapid cooling of molten metal
  - To establish & maintain the arc.
  - Addition of alloying elements.

Arc Welding Electrode

Joining Processes>>

Purpose of Each Ingredient of Flux Coating>>

- **China clay, mica** etc. produce a slag which because of its light weight forms a layer on the molten metal and protects the same from atmospheric contamination.
- Ingredients, like **cellulose, wood, starch,** calcium carbonate etc., form a protective gas shield around the electrode end, arc and weld pool.
- Deoxidizing elements like **Ferro manganese** refine the molten metal.
- Alloying elements like **Ferro alloys of manganese, molybdenum** etc. may be added to impart suitable properties and strength to the weld metal.
- **Iron powder** in the coating improves arc behaviour, bead appearance; helps increase metal deposition rate and arc travel speed.

Safety Devices used in Arc Welding>>

- The welding shield protects the eyes from infra-red & ultra -violet radiations.
- Gloves are used to protect the hands from spark & to insulate from electric shocks
- Chipping hammer & wire brush to clear the slag.
- Apron to protect the clothing from sparks & spatter.
- Earthing clamp will avoid the risk of electric shock.

Fundamentals of Mechanical Engineering

JOINING PROCESSES

Assignment – Unit 2 - 03

<<for Practice>>

1) Explain with neat sketch ….
2) Brief about …..

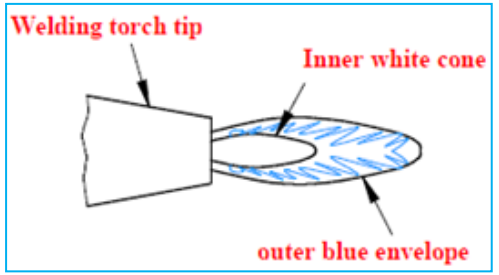
Note:

i) Use new A4 size sheets, provide 1” left and top margin for each sheet.
ii) Write Roll No., Name (at right top), Topic and Assignment No. (at top Middle) in the 1st sheet.
iii) Use red pen to write the questions and blue or black pen for answer
iv) Draw neat sketches using instruments (avoid free hand sketching)

Joining Processes>>

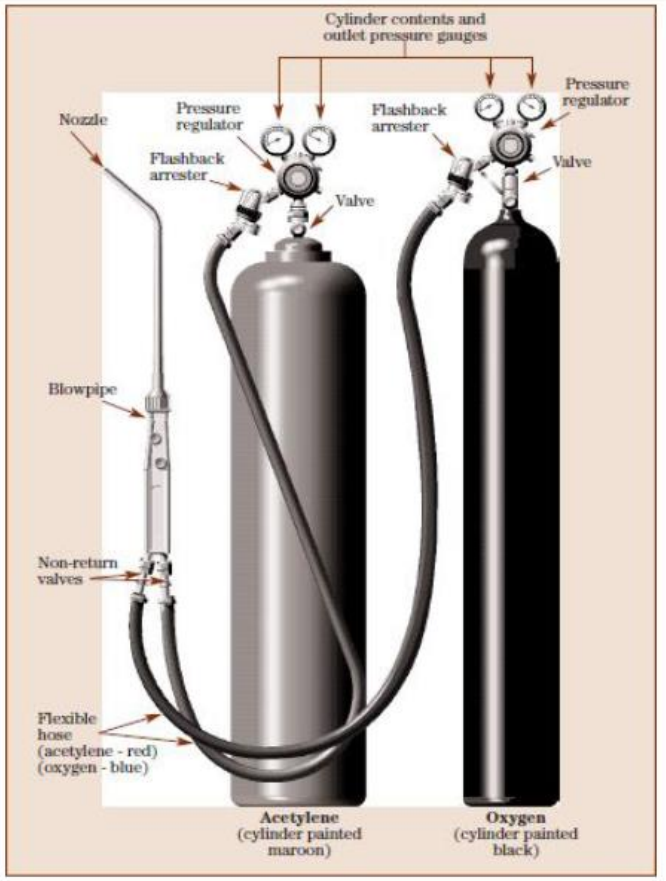
Gas Welding>>

- **Gas welding** is a fusion method of welding in which a strong gas flame is used to raise the temperature of the work piece to melt them. As in arc welding, a filler material is used to fill the joint.
- **The gas combinations** that can be used for heating are;
  - 1. Oxygen & Acetylene
  - 2. Oxygen & Hydrogen



Oxy-Acetylene Welding>>

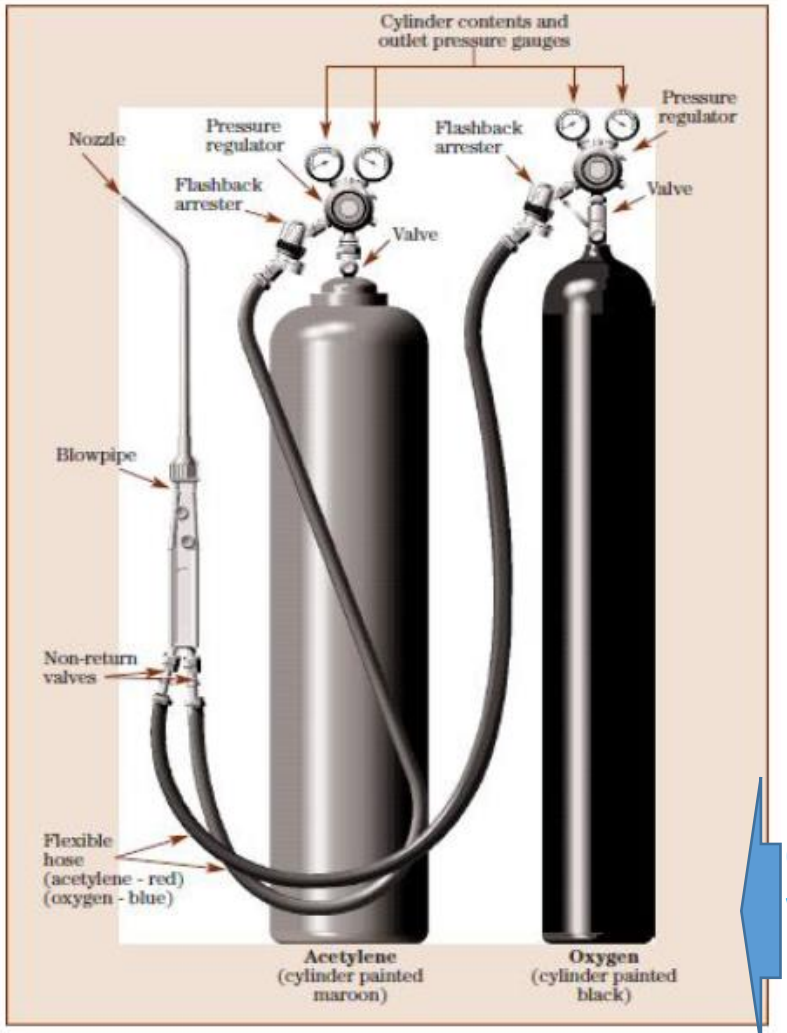
- **When right proportions** of oxygen and acetylene are mixed in the welding torch and when ignited, the flame produced at the nozzle tip is called as the Oxy-Acetylene Flame. This flame when used in welding becomes Oxy-acetylene Welding.
- **The temperature attained** by the Oxy-acetylene flame is around 3200 Centigrade and therefore has the ability to melt all commercial metals. Thus, there is a complete bonding of the joining metals that can be achieved during welding
- **The oxy-acetylene gas equipment** consists of two large steel cylinders; one containing oxygen at high pressure and the other dissolved acetylene also at high pressure.
- **It also consists** of rubber tubes, pressure regulators, & blow torch.
- **The oxygen and acetylene** are supplied to the blow torch separately where both of them get mixed together and come out through the nozzle of the blow torch.



Joining Processes>>

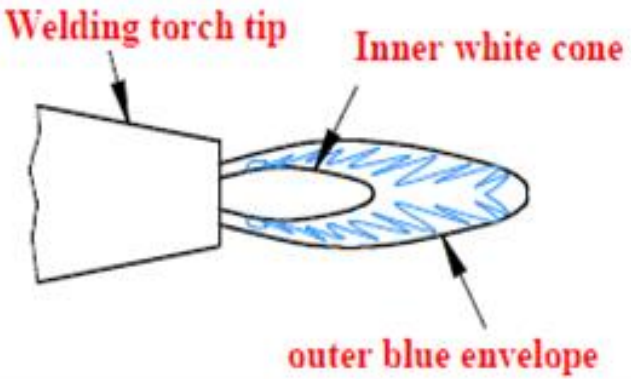
Types of Oxy-Acetylene Flames>

- Neutral Flame
- Carburizing Flame
- Oxidizing Flame

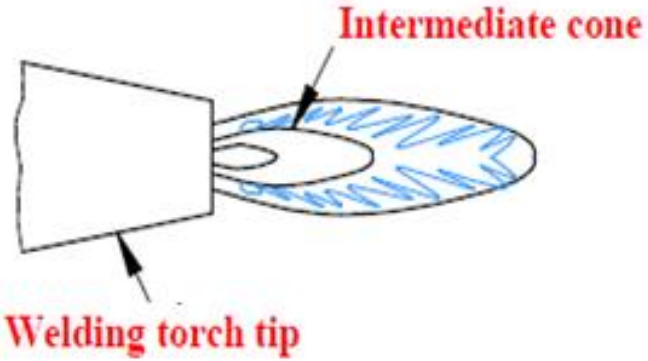


Oxy-Acetylene Welding Facility

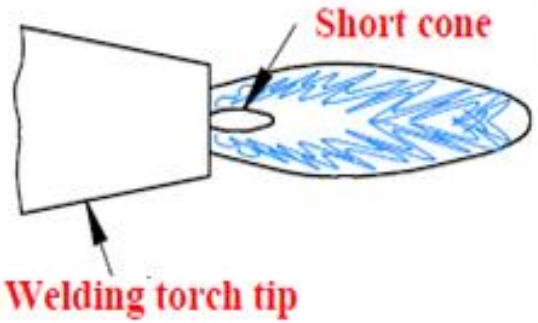
Neutral Flame



Carburizing Flame



Oxidizing Flame



Schematic of Flame

Image of Flame

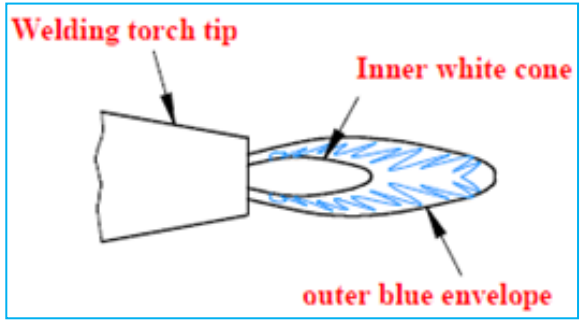
Joining Processes>>

Types of Oxy-Acetylene Flames>

- Neutral Flame
- Carburizing Flame
- Oxidizing Flame

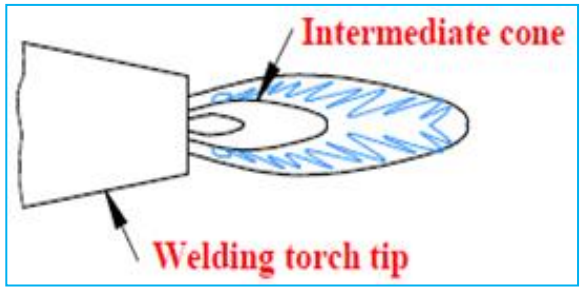
Neutral Flame :

- A Neutral flame is obtained by supplying equal volumes of oxygen & acetylene. (Oxygen: Acetylene = 1: 1)
- A neutral flame consists of an inner small whitish cone surrounded by a sharply defined blue flame.
- Most of the oxy-acetylene welding is done with the neutral flame.



Carburizing Flame :

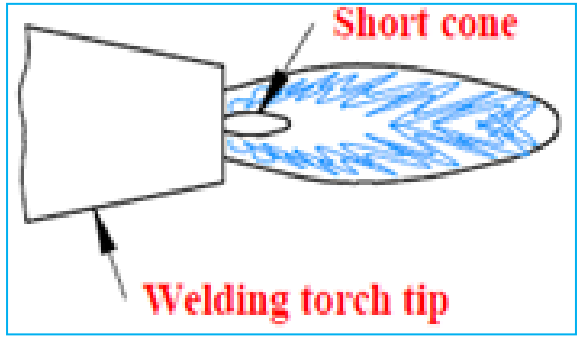
- A Carburizing flame or reducing flame is obtained by supplying excess acetylene. (Oxygen : Acetylene = 0.95 :1)
- It has three cones; an inner white cone, surrounded by an intermediate whitish cone & a bluish envelope flame.
- Carburizing flame is used for welding alloy steels, cast iron & aluminium to protect from oxidizable elements.



Joining Processes>>

Types of Oxy-Acetylene Flames>

Oxidizing Flame:



- The Oxidizing flame is obtained when there is excess oxygen having gas ratio as high as 1.5. (Oxygen: Acetylene = 1.5 : 1)
- In appearance, it resembles a neutral flame except that the inner white cone is somewhat shorter.
- It is used for welding nickel and many non ferrous materials.
- This is used for oxy-acetylene cutting and is not suitable for welding, since the weld metal will be oxidized

Joining Processes>>

Defects in Welding>

Common weld defects include:

- **Lack of fusion:**
  It results from too little heat input and too rapid traverse of the welding torch (gas or electric)
- **Porosity:**
  This occurs when gases are trapped in the solidifying weld metal.
- **Inclusions:**
  These can occur when the slag covering a weld is not totally removed before another weld run.
- **Cracking:**
  This can occur due to thermal shrinkage.
- **Undercut:**
  This is due to excess melting of the parent metal which reduces the strength of the joint.

Joining Processes>>

Soldering > Introduction>

- Soldering is a process in which two or more metal items are joined together by melting and flowing a filler metal (solder) into the joint.
- The filler metal will have a lower melting point than the work piece *(between 1500C-3500C).*
- Soldering differs from welding in that *soldering does not involve melting the work pieces.*
- To clean the surfaces to be joined & to prevent oxidation, Zinc chloride is used as a flux.
- A soldering iron is used to apply heat produced from an electrical source.
- An alloy of lead & tin called 'Soft solder' is used for sheet metal work, plumbing work & electrical junctions. The melting temperature of the soft solder will be between 150 to 350 C.
- An alloy of Copper, Tin and Silver known as Hard solder is used for stronger joints. The soldering temperatures of hard solders ranges from 600 to 900 C.

Soldering Wire

Soldering Flux

Soldering Iron

Joining Processes>>

Soldering >> Tools[1]>

The following are the tools used in soldering:

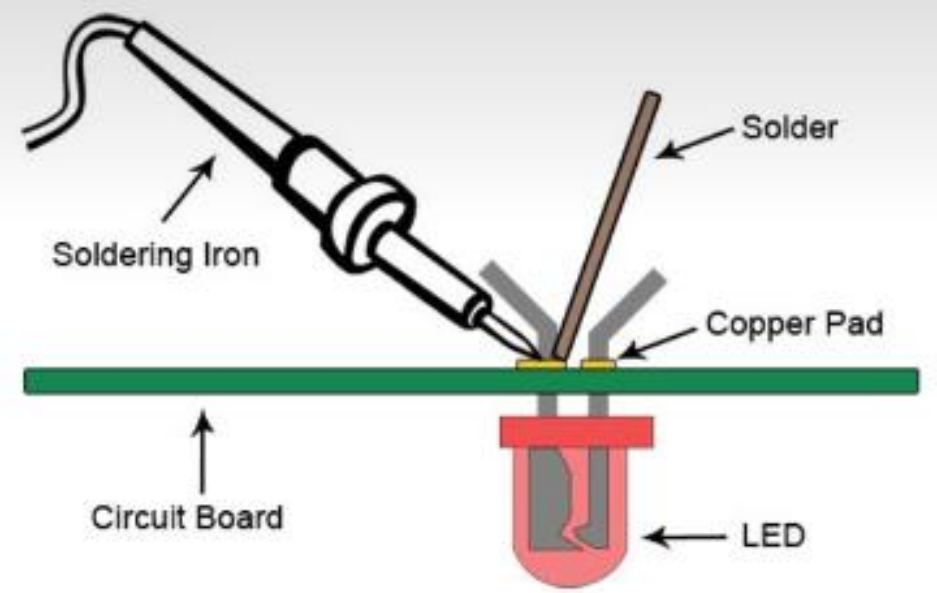
1. Soldering iron
2. Soldering station
3. Iron tips
4. Brass or conventional sponge
5. Soldering iron stand
6. Solder
7. Helping Hand

Helping hand a device consisting of 2 or more clips and sometimes a magnifying glass/light is attached.

Joining Processes>>

Soldering>> Method >

- **Cleaning of Joining surfaces:**
  The joining surfaces are cleaned mechanically to make them free from dust, oil, scales etc and ensure that the molten filler wets the surfaces.
- **Application of flux:**
  The joining surfaces are coated with a flux, usually rosin or borax. This cleans the surfaces chemically and helps the solder in making a bond.
- **Tinning of the surfaces to be soldered:**
  Before carrying out the soldering operation, the soldering iron must be “tinned”. This is to remove a thin that forms on the copper bit.
- **Heating:**
  The soldering iron is then heated and the flowing molten filler metal fills the joint interface. Allow the soldered area to cool and then solidify thus making the joint.
- **Final clean Up :**
  After completing the soldering, and the joints are formed, clean it with steel wool or solvent to remove left –over flux. After this clean the soldering iron using a damp sponge.



Schematic of Soldering [2]

Joining Processes>>

Soldering >> Safety, Advantages and Disadvantages [1]>

Safety in Soldering>

1. The temperature of soldering irons can reach 800 F (Fahrenheit), so it is very important to know where the iron is at all times.
2. When doing soldering, one should use a soldering iron stand to prevent accidental damage.
3. When solder is heated, fumes are released which are very dangerous to the eyes and lungs. It recommends using a fume extractor, a fan with a charcoal filter that absorbs harmful solder fumes.
4. One must wear protective eyewear when doing soldering, to protect from an accidental splash of hot solder.

Advantages of Soldering>

The following are the advantages of soldering:

1. It is a simple and economical process.
2. Since it has been done at relatively low temperatures, no metallurgical damage to the base metal.
3. The soft solder joint can easily be dismantled by simple heating.
4. Operator fatigue is less compared to the welding process. Soldering is divided into two classifications: soft and hard.

Disadvantages of Soldering>

The following are the disadvantages of soldering:

1. The process is limited to the minimum thickness(3mm).
2. A skilled operator is required.
3. The strength of the joint, when compared to welding, is lower.

Extra information...

Joining Processes>>

Soldering >> Applications >

The main applications of soldering are:

1. Soldering is used to join automobile radiator cores.
2. Used to plumbing.
3. Mainly useful in electronic industries like radio, TV, and computers.
4. For joining wires and cables to lugs in electrical industries

Solder is commonly used in:

- Electronics
- Heating
- Air conditioning
- Fire sprinkler
- Radiator manufacturing, repair and
- Sheet metal work

It is also used in

- Jewellery and
- Stained glass work

Extra information….

Many industries, including metalwork and plumbing use soldering techniques [3].

Electronics

Without the use of solder, modern electronics wouldn't be able to function. Solder is a low-melting-point babbitt metal that connects components to substrates and brings electrical continuity so these devices can work perfectly. Printed circuit boards are primarily soldered by using these connections; without them, PCBs would have no way for electricity transfer or signals transmission between parts in the board itself – making it useless!

Metalwork

The world of jewelry-making is often a laborious process requiring precision and patience. One way to make the job easier can be found in soldering, where various pieces are bonded together by heating them up until they fuse into one solid piece. This form of bonding has been used for centuries on jewelry and musical instruments such as violins and trumpets.

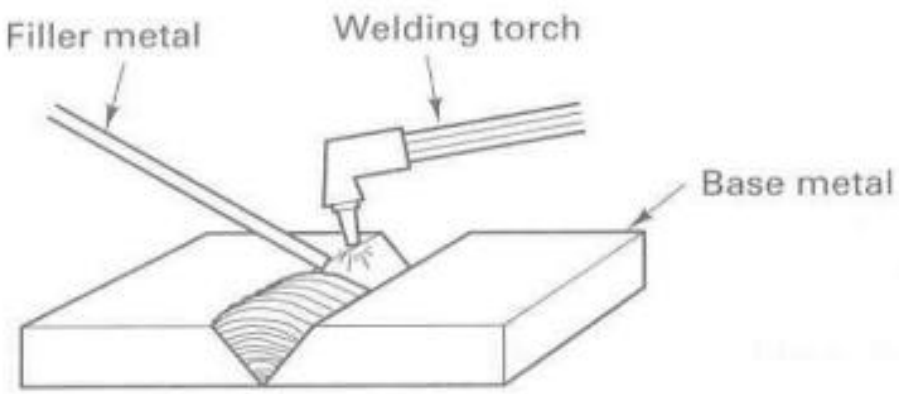
Plumbing

Plumbing technicians use soldering to repair and seal leaky pipes, radiators, fuel tanks, or anything that needs a liquid-tight connection. The most common application is in the water supply line, where heat from an oxyacetylene torch melts solder into the copper joint to be fused with pressure while cooling down.

Joining Processes>>

Brazing>>

- Brazing is akin to soldering in that it also uses filler for joining two metals, albeit a high-temperature one. Similar to welding, brazing creates a mechanical connection between to metal parts[4].
- In a very general sense, brazing is a joining process that relies on the melting, flow, and solidification of a brazing filler metal to form a leak-tight seal, a strong structural bond, or both between materials.
- It is a joining process that produce the coalescence of materials by heating them to the brazing temperature in the presence of a brazing filler metal that has a liquidus temperature above 450ºC and below the solidus temperature of the base materials.
- The process is unique in that this metallurgical bond is formed by melting the brazing filler metal only; the components being joined undergo no melting.
- Brazing is a well-established commercial process capable of producing strong joints.



Schematic of Brazing

Actual process of Brazing

Joining Processes>>

Features of Brazing>>

- **Joining metals by melting filler:** Brazing consists of heating and melting a filler alloy. Once the filler solidifies, the metal pieces are joined.
- **Filler materials:** The filler alloy must have a lower melting temperature than the metal pieces.
- **Dissimilar metals:** Brazing allows for the joining of different metals like aluminum, silver, copper, gold, and nickel.
- **Flux is often used**: Liquid flux material promotes wetting of the metal parts, which lets the filler flow over them, so they can be joined more easily.
- **Special fluxes** are used (Borax) to: remove the oxide improve the fluidity of the fillers wet the joint surfaces
- **Strength**: While properly brazed joins can be stronger than the individual pieces, they are not as strong as welded joints.
- **Brazing Materials:** copper alloys silver alloys aluminium alloys

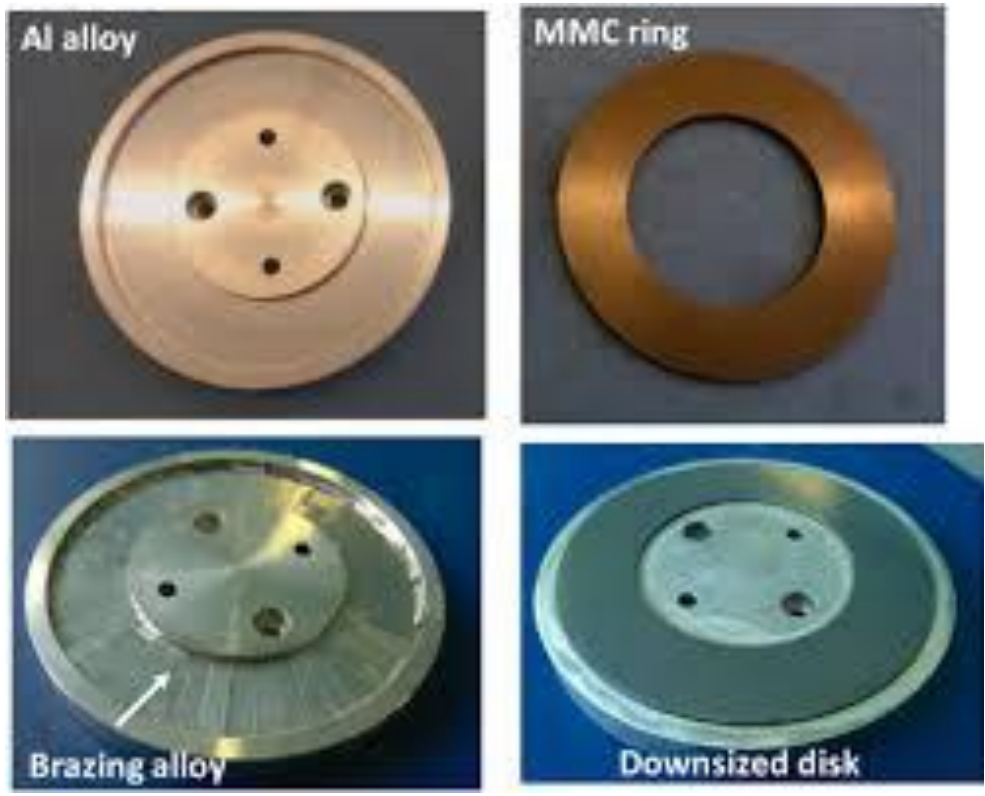
Advantages, Uses / Applications of Brazing>>

- It is widely used in industry because, in large part, it is capable of joining most metallic and ceramic materials.
- It is a versatile process that can be performed using manual techniques as well as automated production modes.
- Brazing lends itself to the production of large assemblies and assemblies composed of dissimilar metals.
- Brazing produces a tiny, clean fillet in contrast to the irregular bead made by welding, an advantage when appearance is critical.
- One of the main advantages of brazing is usually associated with cost savings.
- High production processes adapt well to today's improved processes.
- Brazing especially adapts to large production quantities as well as single individual quantities.

Joining Processes>>

Applications of Brazing>>

Copper - Copper



Alloys

Joining Processes>>

Differences between Brazing, welding and soldering [4]>>

Welding, soldering, and brazing are used for different jobs.
Depending on the work to be done, one has to consider factors like metal type, desired strength, and ultimate purpose to determine how to join two or more pieces of metal.

1. Strength

- **Welding** joints are the strongest joints used to bear the load.
  Strength of the welded portion of joint is usually more than the strength of base metal.
- **Soldering** joints are weakest joints out of three.
  Not meant to bear the load.
  Use to make electrical contacts generally.
- **Brazing** joints are weaker than welding joints but stronger than soldering joints.
  This can be used to bear the load up to some extent.

2. Temperature

- **Welding**: Temperature required is 3800°C in welding joints.
- **Soldering**: Temperature requirement is up to 450°C in soldering joints.
- **Brazing**: Temperature may go to 600°C in brazing joints.

Differences between Brazing, welding and soldering [4]>> Contd…

3. Heating of Work Pieces

- **Welding**: To join work pieces need to be heated till their melting point.
- **Soldering**: Heating of the work pieces is not required.
- **Brazing**: Work pieces are heated but below their melting point.

4. Change in Mechanical Properties

- **Welding** : Mechanical properties of base metal may change at the joint due to heating and cooling.
- **Soldering** : No change in mechanical properties after joining.
- **Brazing** : May change in mechanical properties of joint but it is almost negligible.

5. Heat Treatment

- **Welding** : Heat treatment is generally required to eliminate undesirable effects of welding.
- **Soldering** : No heat treatment is required.
- **Brazing** : No heat treatment is required after brazing.

6. Preheating of Workpiece

- **Welding** : No preheating of workpiece is required before welding as it is carried out at high temperature.
- **Soldering** : Preheating of workpieces before soldering is good for making good quality joint.
- **Brazing** : Preheating is desirable to make strong joint as brazing is carried out at relatively low temperature.